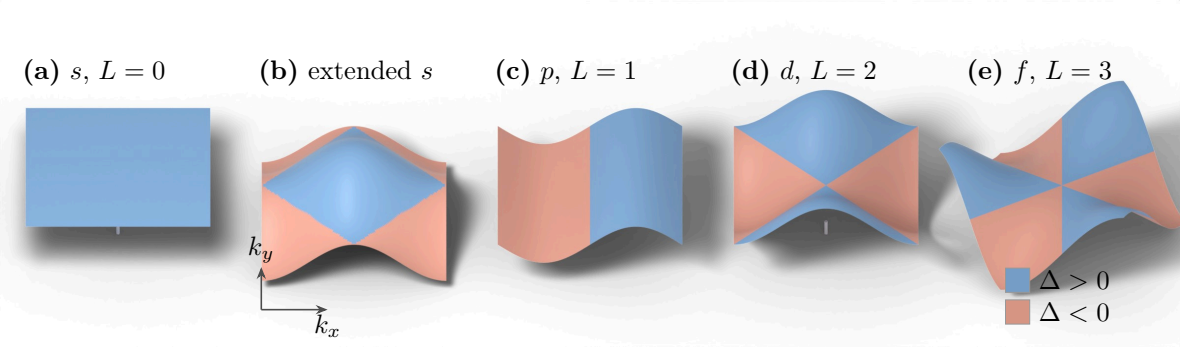

(a) s, $L = 0$
(b) extended s
(c) p, $L = 1$
(d) d, $L = 2$
(e) f, $L = 3$
k_y
k_x
$\Delta > 0$
$\Delta < 0$